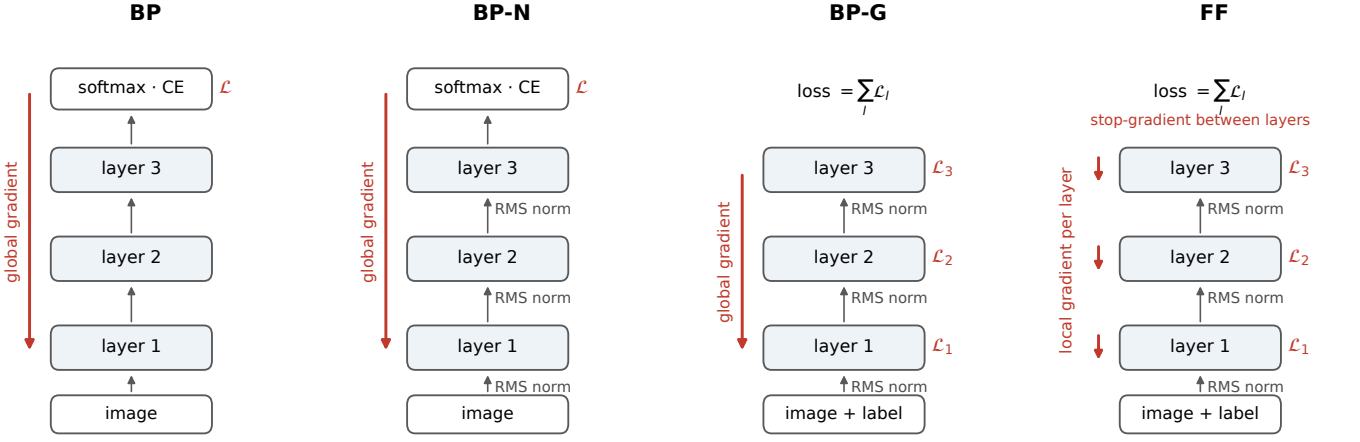
BP
softmax · CE
$\mathcal{L}$
layer 3
layer 2
layer 1
image
global gradient
BP-N
softmax · CE
$\mathcal{L}$
layer 3
RMS norm
layer 2
RMS norm
layer 1
RMS norm
image
global gradient
BP-G
loss $= \sum_l \mathcal{L}_l$
layer 3
$\mathcal{L}_3$
RMS norm
layer 2
$\mathcal{L}_2$
RMS norm
layer 1
$\mathcal{L}_1$
RMS norm
image + label
global gradient
FF
loss $= \sum_l \mathcal{L}_l$
stop-gradient between layers
layer 3
$\mathcal{L}_3$
RMS norm
layer 2
$\mathcal{L}_2$
RMS norm
layer 1
$\mathcal{L}_1$
RMS norm
image + label
local gradient per layer